

Homework 2: Rates, Ratios, and Discounting

Algori QI Education | Data downloaded 22 September 2026

1. Present value

present_value([10, 15, 20], 0.10) = 36.51. Each cash flow is discounted by $(1 + \text{rate})$ raised to its year number.

2. Bond priced at its coupon rate

bond_price(1000, 0.04, 10, 0.04) = \$1,000.00. The final cash flow includes both the \$40 coupon and the \$1,000 face value.

3. Bond price and market rate

Market rate	Bond price
2.00%	\$1,179.65
4.00%	\$1,000.00
4.96%	\$925.73



The curve slopes downward and is convex: a given fall in rates increases the price more than an equal rise in rates decreases it.

4-6. Beta from five years of daily closing prices

Selected sectors: AAPL (technology), JPM (financials), and XOM (energy). Each series, including SPY, contains 1,254 trading days. Annualised return uses compounded daily growth; volatility is the standard deviation of daily returns multiplied by the square root of 252; beta is calculated from daily returns against SPY.

Ticker	Annualised return	Annualised volatility	Beta vs SPY
AAPL	18.38%	28.05%	1.16
JPM	16.20%	24.45%	0.87
XOM	22.83%	26.68%	0.39

Ranked by beta: **AAPL > JPM > XOM**.

Ranked by volatility: **AAPL > XOM > JPM**.

7. Interpretation

The rankings do not fully agree: JPM ranks above XOM on beta, while XOM ranks above JPM on volatility. XOM is the clearest example: its annualised volatility is 26.68%, yet its beta is only 0.39. That combination means much of XOM's movement over this period was company- or energy-sector-specific rather than movement shared with the broad equity market. In other words, XOM was relatively jumpy, but its jumps were not especially sensitive to SPY, so diversification can reduce more of its risk than it can for a high-beta stock.